

IILM UNIVERSITY, GREATER NOIDA

Data Analytics with Python & Power BI

AICTE – EduSkills Virtual Internship Program

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Organization: EduSkills (AICTE Virtual Internship)



What This Presentation Covers



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About the Internship



Program

AICTE – EduSkills Virtual Internship Program

Domain

Data Analytics with Python & Power BI

Duration

June 2026 – Aug 2026 (8 weeks)

Mode

Virtual / Online, self-paced with weekly milestones

Evaluation

Weekly assessments, project submissions & Final Assessment Test

Objectives



Build strong foundations in Python programming for data analytics



Learn SQL for querying, cleaning, and transforming structured data



Gain hands-on expertise in Microsoft Power BI for dashboarding



Integrate Python, SQL, and Power BI into one analytics workflow



Apply all skills to a real-world style business analytics project

Scope of the Project

In Scope

- ✓ Data cleaning & preprocessing with Python
- ✓ SQL-based data extraction and analysis
- ✓ Power BI dashboard design & publishing
- ✓ Capstone: e-commerce dataset analytics

Out of Scope

- ✗ Production-grade software deployment
- ✗ Real-time / streaming data pipelines
- ✗ Integration with live enterprise systems
- ✗ Enterprise-scale data governance

Week-wise Learning Journey (1-4)

01

Data Analytics Foundations & Python Basics

Environment setup, core Python, NumPy & Pandas fundamentals

02

Data Cleaning & SQL Fundamentals

Pandas preprocessing; SQL SELECT, WHERE, GROUP BY, joins

03

Advanced SQL & Power BI Basics

Subqueries, CTEs, window functions; Power BI intro & basic visuals

04

Python-SQL Integration & Data Modeling

DB connections, automation; Power BI modeling & DAX basics

Week-wise Learning Journey (5–8)

05

Advanced Visualization

Drill-through, bookmarks, filters; Matplotlib & Seaborn EDA

06

Integration, Publishing & Collaboration

Python-in-Power BI; publishing to Power BI Service

07

Real-World Projects & Best Practices

Sales performance analysis; customer segmentation

08

Capstone Project

E-commerce data prep, SQL analysis, dashboard delivery

Tools & Technologies Used



Python

Core programming for cleaning & scripting



Pandas

Data wrangling & transformation



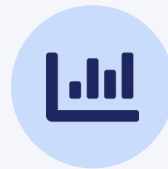
NumPy

Numerical computation



SQL

Querying, joins, CTEs, window functions



Power BI

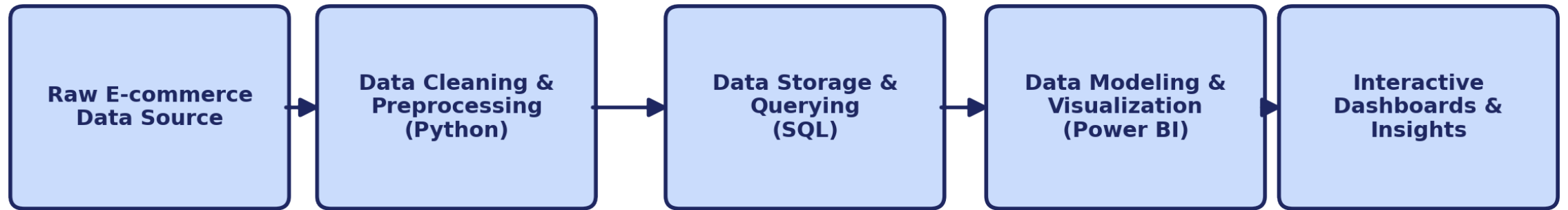
Data modeling, DAX, dashboards



Jupyter

Interactive analysis environment

System Architecture



Raw data flows from ingestion through Python-based cleaning, SQL-based querying, and Power BI modeling into decision-ready interactive dashboards.

Capstone Project

E-commerce Dataset: Data Preparation, SQL Analysis & Power BI Dashboard



Learning Outcomes

- ✓ Hands-on proficiency in the Python-SQL-Power BI analytics stack
- ✓ Competence writing SQL queries for real-world data tasks
- ✓ Stronger structured problem-solving and workflow discipline
- ✓ Ability to convert raw, messy data into analysis-ready datasets
- ✓ Skill designing and publishing decision-ready dashboards

Conclusion

The AICTE-EduSkills Virtual Internship on Data Analytics with Python & Power BI provided structured, hands-on exposure to the complete analytics workflow — from data cleaning and SQL analysis to building professional Power BI dashboards — strengthening both technical skill and confidence in solving real-world business problems.

Thank You

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